Algorithm 1 vAttention

Require: KVCache $\mathbf{K}, \mathbf{V}$: $n \times d$, $q : 1 \times d$. Parameters $f_s, f_l, f_t \in (0,1)$: fraction of total tokens for sink, sliding window and top-k tokens. Adaptive sampling parameters $f_b \in (0,1)$: base sampling rate, $\varepsilon, \delta \in (0,1)$ user specified guarantee.

1: $\mathcal{I}_s \leftarrow \{0, 1, \ldots, \lfloor f_s n \rfloor - 1\}$
2: $\mathcal{I}_l \leftarrow \{n - \lfloor f_l n \rfloor, \ldots, n-1\}$
3: $\mathcal{I}_t \leftarrow \texttt{pred-top-index}(f_t, n, K, V, q)$
4: $\mathcal{I}_{\text{f}} \leftarrow \mathcal{I}_s \cup \mathcal{I}_l \cup \mathcal{I}_t$, $\mathbf{p}_{\text{f}} = \mathbf{1}^{|\mathcal{I}_f|}$
5: $n_s \leftarrow n - |\mathcal{I}_f|$
6: $b \leftarrow \texttt{budget}(\mathcal{I}_{\text{f}}, f_b, n_s, \varepsilon, \delta, \mathbf{K}, \mathbf{V}, q)$
7: $\mathcal{I}_{\text{dyn}} \leftarrow \texttt{uniform-sample}(\mathcal{I}_f, b, n)$
8: $\mathbf{p}_{\text{dyn}} \leftarrow \frac{b}{n_s} \cdot \mathbf{1}^{|\mathcal{I}_{\text{dynamic}}|}$
9: $S \leftarrow [\mathcal{I}_{\text{f}}, \mathcal{I}_{\text{dynamic}}]$
10: $p \leftarrow [\mathbf{p}_{\text{f}}, \mathbf{p}_{\text{dynamic}}]$
11: **return** $\text{SDPA}_{S,p}(\mathbf{K}, \mathbf{V}, q)$

Algorithm 2 `budget`

Require: KVCache $\mathbf{K}, \mathbf{V}$: $n \times d$, $q : 1 \times d$, Parameters: n_s sampling range, $f_b \in (0,1)$: base sampling rate, $\epsilon, \delta \in (0,1)$ probabilistic guarantee required by user, $\mathcal{I}_f$ static index.

1: $\mathcal{I}_{bs} \leftarrow \texttt{uniform-sample}(\mathcal{I}_f, f_b, n)$
2: $\hat{\Sigma}, \hat{\sigma}, ||\hat{N}||, \hat{D} \leftarrow$ $\texttt{compute-stats}(\mathcal{I}_f, \mathcal{I}_{bs}, K, V, q)$
3: $b = b_{\mathbf{SDPA}}(\epsilon, \delta, \hat{\mathbf{Tr}}(\Sigma), \hat{\sigma}, ||\hat{N}||_2, \hat{D}, K, V, q)$
4: **return** b

does not have significant outliers (i.e., when residual attention scores have comparable magnitudes). Let $\mathcal{I}_f = \mathcal{I}_s \cup \mathcal{I}_l \cup \mathcal{I}_t$ and $|\mathcal{I}_s| + |\mathcal{I}_l| + |\mathcal{I}_t| = n_f$.

Let $n_s{=}n{-}n_f$ denote the number of residual indices. vAttention uniformly samples indices $\mathcal{I}_{dyn}$ from residual indices. The set of indices and associated sampling probabilities can be expressed as,

$$S = \mathcal{I}_f \cup \mathcal{I}_{dyn} \qquad P_i = \frac{|\mathcal{I}_{dyn}|}{n_s} \text{ if } i \in \mathcal{I}_{dyn} \text{ and 1 otherwise} \tag{4}$$

Then the vAttention computation can be written as, vAttention$(K, V, q) = \text{SDPA}_{S,P}(K, V, q)$. More explicitly, we can write vAttention(K, V, q) as

$$\frac{N}{D} = \frac{N_f + N_{dyn}}{D_f + D_{dyn}} = \frac{\sum_{i \in \mathcal{I}_f} (\exp\langle K[i], q\rangle V[i]) + \frac{n_s}{|\mathcal{I}_{dyn}|} \sum_{j \in \mathcal{I}_{dyn}} (\exp\langle K[j], q\rangle V[j])}{\sum_{i \in \mathcal{I}_f} (\exp\langle K[i], q\rangle) + \frac{n_s}{|\mathcal{I}_{dyn}|} \sum_{j \in \mathcal{I}_{dyn}} (\exp\langle K[j], q\rangle)} \tag{5}$$

where N_f (alt. D_f) comes from deterministic indices and N_{dyn} (alt. D_{dyn}) comes from stochastic indices that estimate the rest of the numerator (alt. denominator).

The theoretical guarantees in vAttention arise from the careful choice of the sample size, i.e., $|\mathcal{I}_{dyn}|$. The sample size is selected to ensure that the attention approximation is ϵ-relative accurate with probability $1 - \delta$. We now explain how the budget is chosen. The choice of sample size is guided by the following result on estimating the sum of n quantities (See Appendix D for proof)

Lemma 4.1 (Estimating vector sum). *Let $\mathbf{s} = \sum_{i=1}^{n_s} \mathbf{r}_i, \mathbf{s} \in R^d$ be a sum of n_s vector quantities $\mathbf{r}_i \in R^d \,\forall i$ which has to be estimated using a sample $\mathcal{I}_b$ of size b. Let Σ be the covariance matrix for the population $\{\mathbf{r}_i\}_{i=1}^{n_s}$. Let $\hat{\mathbf{s}}_b = \frac{n_s}{b}\left(\sum_{i \in \mathcal{I}_b} \mathbf{r}_i\right)$ be the estimate. Let Φ be the CDF for the normal distribution. Then for a large enough b if,*

$$b \geq \left(\Phi^{-1}\left(1 - \frac{\delta}{2}\right) \frac{n_s \sqrt{\mathbf{Tr}(\Sigma)}}{\tau}\right)^2 \quad \textit{then} \quad \mathbf{Pr}(||\hat{\mathbf{s}} - \mathbf{s}||_2 > \tau) \leq \delta \tag{6}$$

for any arbitrary $\tau \in R$ and $\delta \in (0,1)$.

Comment We leverage the Central Limit Theorem (CLT) to approximate the sum using a sufficiently large sample. We can obtain a similar result for scalar quantities by setting $d = 1$ in the theorem above. Detailed empirical analysis of using optimistic CLT and conservative Hoeffding's method for budget computation is presented in E.

We can use the above lemma to compute budget required for (ϵ, δ) approximations of the numerator and denominator independently as mentioned in Corollary D.2 D.3. These are obtained by setting $\tau =$